

Vision Playground

A collection of interactive eye training exercises · priyanka-mondal.github.io/vision-playground

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About This Project

I have been dealing with **double vision (diplopia)** since childhood. It is something I have lived with for as long as I can remember — the world sometimes appears as two overlapping images, making everyday tasks like reading, driving, and screen time more tiring than they should be.

Over the years I have gone through **vision therapy** and tried many different eye exercises — some recommended by my vision doctor, others I discovered on my own through research. Some of them genuinely helped. They improved my ability to fuse images, increased my convergence range, and reduced the fatigue I felt after long hours of work.

To make things easier for myself — and for others who might be in a similar situation — I built **Vision Playground**. It brings all the science-backed exercises I found useful into one place: free, browser-based, and always available. No appointments, no expensive equipment, no subscriptions.

The only prop you will ever need is a pair of **red-green anaglyph glasses** — the kind that cost a few dollars on Amazon.com. These unlock the more advanced RDS (Random Dot Stereogram) exercises, which are specifically designed to train binocular fusion and stereoscopic depth perception. If you are just starting out, begin with the no-glasses exercises (the website is at [Vision Playground](https://priyanka-mondal.github.io/vision-playground)) and build up from there. Get the glasses only when you feel ready to move to the advanced levels.

I hope this helps you as much as it has helped me.

■■ RDS Stereogram Exercises (Red-Green Glasses Required)

These exercises use **red-green anaglyph glasses** (red lens on the left eye, green lens on the right eye). They train binocular vision, vergence, and stereoscopic depth perception. The random-dot patterns are generated fresh every 4 seconds so you cannot memorise the image.

■ Position Game

A random-dot stereogram fills a large circle. Hidden inside is a smaller circle that appears to float at a different depth. The hidden circle is placed in one of four positions — top, bottom, left, or right — relative to the centre of the fused image. You have 30 seconds to score as many correct identifications as possible.

Controls:

- $\uparrow \downarrow \leftarrow \rightarrow$ — identify where the 3D circle appears to float
- Start Game button — begin a new 30-second round

Goal:

Score as many correct answers as possible within 30 seconds. Each correct answer adds 1 point and a green flash confirms it.

■ *Keep both eyes open and relaxed. The hidden circle pops out in depth once your visual system finds the binocular match — this can take a few seconds.*

■ Convergence Test

Two large random-dot circles — one rendered in red (left eye) and one in green (right eye) — start overlapping and slowly drift apart by 0.5 prism diopters (Δ) every few seconds. While wearing your anaglyph glasses, fuse the two circles into one by adjusting your eye vergence. Press SPACE the moment you can no longer hold fusion. The app records your maximum convergence amplitude.

Controls:

- SPACE — signal that fusion is lost; records the current diopter level
- Start button — begin a new convergence measurement session

Goal:

Push your binocular fusion as far as possible. Your maximum fused diopter is saved so you can track improvement over multiple sessions.

■ *Relax your eyes rather than straining — forced vergence fatigues faster. Try again after a short rest to see if your range improves.*

■ Combined Test

The most demanding exercise. The two circles drift apart by 0.5 Δ every 10 seconds (increasing vergence demand), while a hidden 3D circle is simultaneously present inside each circle at the same relative position. You must maintain fusion of the two big circles AND identify the position of the hidden element within each 10-second window. A correct answer immediately advances to the next diopter level. The dot pattern refreshes every 4 seconds to keep the task challenging.

Controls:

- $\uparrow \downarrow \leftarrow \rightarrow$ — identify the hidden circle's position (top / bottom / left / right)
- SPACE — signal fusion lost; ends the session and shows your results
- Start button — begin at 0 Δ and advance automatically

Goal:

Reach the highest diopter level while maintaining fusion. Your score (correct answers) and max fused diopter are displayed after each session.

■ *The hidden circle is at the same quadrant position within each big circle. When fused correctly, it appears clearly offset from centre in one direction. Use the faint arrow hints on screen as a spatial reference.*

■ General Eye Training Exercises

These exercises require no special glasses and can be done anytime to reduce eye strain, improve eye muscle control, and build visual stamina.

Smooth Pursuit

A dot moves smoothly around the screen in a circular or figure-eight path. Track it with your eyes while keeping your head completely still.

Controls:

- Start / Stop button — begin or pause the exercise

Goal:

Maintain smooth, continuous eye movement that exactly follows the dot. Avoid jerky catch-up movements.

■ *If you find yourself moving your head, place your chin on your fist as a reminder.*

Saccade Training

Multiple targets appear on screen and one is highlighted at a time. Shift your gaze as quickly and accurately as possible to each newly highlighted target.

Controls:

- Start / Stop button — begin a session

Goal:

Minimise reaction time between target changes. Saccades are the fast eye movements used in reading and scene scanning.

■ *Look directly at the centre of each target rather than the surrounding area.*

Figure 8

Trace an imaginary figure-eight (∞) path with your eyes, following the on-screen guide. Keep your head still throughout.

Controls:

- Start / Stop button

Goal:

Complete smooth, symmetrical loops in both directions, training all six extra-ocular muscles.

■ *Go slowly at first. Speed is less important than keeping the path smooth and continuous.*

Focus Shift

Alternate between focusing on a near dot (close to your face) and a far text target on the screen. Hold each focus for 2 seconds before switching.

Controls:

- Start button — begin the timed alternation cycle

Goal:

Train the ciliary muscles that control lens accommodation (focus adjustment). Particularly useful for people who spend long hours looking at screens.

■ *Hold a finger close to your nose as the near target if the on-screen dot is too far.*

20-20-20 Timer

A guided countdown reminder for the 20-20-20 rule: every 20 minutes of screen time, look at something at least 20 feet away for 20 seconds.

Controls:

- Start button — begin the 20-minute work countdown

Goal:

Reduce digital eye strain by giving your accommodation system regular breaks.

■ *Set this running whenever you start a work session and follow every rest prompt.*

Palming Relaxation

Cup your palms gently over your closed eyes without pressing on them, blocking all light. Breathe slowly and deeply, and let your eyes rest in complete darkness.

Controls:

- Start button — begins a timed relaxation session

Goal:

Release tension in the eye muscles and the visual cortex. Best done after any of the more demanding exercises.

■ *Visualise a peaceful dark space rather than trying to see anything. Even 2 minutes of palming can noticeably reduce eye fatigue.*

■ Convergence Rings (Red-Green Glasses Required)

Convergence Rings

Two outline rings are displayed — one red, one green — at a fixed separation. Wear your red-green anaglyph glasses and adjust your eye vergence until the two rings fuse into a single ring. A hidden number appears inside when you achieve correct fusion.

Controls:

- Visual only — adjust vergence by relaxing or converging your eyes

Goal:

Successfully fuse the two rings and read the hidden number. Use as a quick binocular check before the more demanding RDS sessions.

■ *Start by looking slightly beyond the screen (diverge) rather than crossing your eyes. The number will snap into focus once fusion is achieved.*